

Cores of the Merged Johnson Graphs

$$J(2k + 1, k)_{\{2, \dots, k-2\}}$$

Jihun Kim*

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Abstract

For $k \geq 4$, let G_k be the graph on the k -subsets of $[2k + 1]$, where two distinct vertices are adjacent when their intersection has size between 2 and $k - 2$. We prove that

$$\text{End}(G_k) = \text{Aut}(G_k) \cong S_{2k+1},$$

and hence that G_k is a core. We also show that $\alpha(G_k) = 2k + 3$ and classify the maximum independent sets. The proof reduces a nontrivial core retraction to a translate-transversal identity and then uses the multiplicity-free Johnson decomposition to separate the spectral supports of a fibre and the core image. The resulting support restrictions would make one of the remaining Johnson relation counts negative, a contradiction. It follows that the almost-uniform 3-qualitative independence hypergraph $3\text{-AUQI}(n, 2)$ is a core for every odd $n \geq 9$.

Keywords. Graph core; graph endomorphism; merged Johnson graph; Johnson association scheme; qualitative independence; extremal set system.

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1 Introduction

A finite graph is a *core* if every endomorphism is an automorphism; equivalently, it has no retraction onto a proper subgraph homomorphically equivalent to it [8, 7]. Determining cores is particularly subtle for highly symmetric graphs: the automorphism group describes the invertible maps but does not, by itself, exclude noninvertible foldings. General results for symmetric and geometric graphs give strong restrictions on their cores [3, 5, 10], but do not decide the merged relations considered here.

For $k \geq 4$, define

$$\Omega = \binom{[2k + 1]}{k}, \quad V(G_k) = \Omega, \quad A \sim B \iff 2 \leq |A \cap B| \leq k - 2. \quad (1)$$

If $d(A, B) = k - |A \cap B|$ is Johnson distance, then the edge set is the union of relations $2, \dots, k - 2$. The same interval occurs in both intersection and distance notation; thus

$$G_k = J(2k + 1, k)_{\{2, \dots, k-2\}}.$$

*Email: jihunkimkw@gmail.com

Jones's classification of automorphisms of merged Johnson graphs gives

$$\text{Aut}(G_k) \cong S_{2k+1} \quad (2)$$

in its natural action on Ω [9, Theorem 2]. Indeed, for the Johnson distance set $I = \{2, \dots, k-2\}$, the relevant exceptional condition $I = k+1-I$ does not occur, since $k+1-I = \{3, \dots, k-1\} \neq I$. The remaining question for this family is whether G_k admits a noninjective endomorphism.

The family arises in qualitative independence. For odd $n = 2k+1$, the vertices of $3\text{-AUQI}(n, 2)$ are almost-balanced binary strings, represented by k -subsets of $[n]$, and a hyperedge is a triple realizing all eight binary patterns. Thomas and Akhtar identify G_k as the relevant 2-section for $3\text{-AUQI}(2k+1, 2)$ and show that if this merged Johnson graph is a core, then so is the hypergraph [12, Lemma 7 and Proposition 7]. They pose the odd cases as a direction for further work. Theorem 1.1 resolves that question for every odd $n \geq 9$.

Theorem 1.1. *For every integer $k \geq 4$,*

$$\text{End}(G_k) = \text{Aut}(G_k) \cong S_{2k+1}.$$

In particular, G_k is a core.

The proof begins with the extremal and divisibility constraints on the fibres of a minimum-rank retraction. Each fibre meets every translate of the core image in exactly one vertex. In the language of permutation groups, this is a non-separation witness for the natural S_{2k+1} -action; see [1]. The additional fact that the fibre is independent restricts its distance distribution to three Johnson relations. The transversal identity separates the nonconstant Johnson supports of the fibre and the core image, after which positivity excludes first the proper divisors of $2k+3$ and then the remaining fibre size $2k+3$.

We also characterize the maximum independent sets, although only the value $\alpha(G_k) = 2k+3$ is used in the core argument.

2 The extremal independent families

An independent family $\mathcal{F} \subseteq \Omega$ has

$$|A \cap B| \in \{0, 1, k-1\} \quad (A \neq B \in \mathcal{F}). \quad (3)$$

General restricted-intersection results are given in [4]. In the case $n = 2k+1$ and $L = \{0, 1, k-1\}$, the following proposition gives the sharp bound and characterizes equality.

Proposition 2.1. *For every $k \geq 4$,*

$$\alpha(G_k) = 2k+3.$$

Moreover, every maximum independent family is, up to a permutation of $[2k+1]$, of the following form. Choose disjoint sets $U, T, \{p\}$ with $|U| = k+1$ and $|T| = k-1$; then

$$\mathcal{M}(U, T, p) = \binom{U}{k} \cup \{T \cup \{x\} : x \in U \cup \{p\}\}. \quad (4)$$

Consequently the number of maximum independent families is

$$(k+2) \binom{2k+1}{k-1}.$$

Proof. Define $A \equiv B$ when $A = B$ or $|A \cap B| = k - 1$. If $A \equiv B \equiv C$, then $|A \cap C| \geq k - 2 \geq 2$; hence (3) shows that $A \equiv C$. Thus the $\equiv$ -classes are cliques in the ordinary Johnson graph. Each such clique is contained in one of the two familiar types

$$\begin{aligned} \text{star:} \quad & \{T \cup \{x\} : x \in R\}, \quad |T| = k - 1, \quad R \subseteq [2k + 1] \setminus T, \\ \text{top:} \quad & \{U \setminus \{u\} : u \in R\}, \quad |U| = k + 1, \quad R \subseteq U. \end{aligned}$$

Indeed, after fixing two members with a common $(k - 1)$ -set, every third member either contains that common set or is contained in their $(k + 1)$ -element union.

Suppose first that a star component has size at least four. An outside block must be disjoint from its common $(k - 1)$ -set T . Otherwise it contains at most one point of T and must avoid every point indexing the star component. If that component has size $r \geq 4$, at most $1 + (k + 2 - r) < k$ points remain available. Therefore all outside blocks lie in the $(k + 2)$ -set $W = [2k + 1] \setminus T$. Two distinct outside blocks meet in at least $k - 2$ points, while independence restricts their intersection to $\{0, 1, k - 1\}$. Since $k \geq 4$, they must meet in $k - 1$ points. Thus all outside blocks form one Johnson clique of size at most $k + 1$. The original star has size at most $k + 2$, so $|\mathcal{F}| \leq 2k + 3$.

Suppose next that a top component has size at least four, contained in $\binom{U}{k}$ with $|U| = k + 1$. Put $W = [2k + 1] \setminus U$, so $|W| = k$. For an outside block B , write $a = |B \cap U|$. Comparison with $U \setminus \{u\}$ gives $a - \mathbf{1}_{\{u \in B\}} \leq 1$ for each index u of the component. If $a \geq 3$, then $a - \mathbf{1}_{\{u \in B\}} \geq 2$, a contradiction. If $a = 2$, the inequality forces all four component indices to lie in B , contradicting $|B \cap U| = 2$. Hence $a \leq 1$, and B is either W or has the form

$$\{u\} \cup (W \setminus \{w\}), \quad u \in U, \quad w \in W.$$

Two blocks of the latter type are compatible only when they have the same u or the same w . Thus they lie in a row or a column of the complete bipartite grid $U \times W$; with W included, their number is at most $k + 2$. The top has size at most $k + 1$, again giving $2k + 3$.

It remains to treat families whose components all have size at most three. Distinct components have disjoint 2-shadows, since a common pair would give intersection at least two. A component of size $r = 1, 2, 3$ contains at least, respectively,

$$\binom{k}{2}, \quad \frac{(k - 1)(k + 2)}{2}, \quad \frac{k(k + 1)}{2}$$

distinct pairs. Each quantity is at least $rk(k + 1)/6$. Summing disjoint shadows gives

$$|\mathcal{F}| \frac{k(k + 1)}{6} \leq \binom{2k + 1}{2} = k(2k + 1),$$

and therefore

$$|\mathcal{F}| \leq \frac{6(2k + 1)}{k + 1} < 2k + 3.$$

The strict inequality follows from

$$(2k + 3)(k + 1) - 6(2k + 1) = 2k^2 - 7k - 3 > 0 \quad (k \geq 4).$$

The family in (4) has size $(k + 1) + (k + 2)$ and satisfies (3), proving the value of $\alpha(G_k)$. Equality in either large-component case forces both component bounds to be tight. In the star case, the outside k -sets in the $(k + 2)$ -set W correspond, by complementation, to a maximum pairwise-intersecting family of 2-subsets of W . Since $|W| = k + 2 \geq 6$, a maximum such family is a full star, and hence

a full top on a $(k + 1)$ -subset $U \subset W$. The one remaining point is p . In the top case, equality forces the full top and $k + 2$ outside blocks. A row of $U \times W$, together with W , has only $k + 1$ blocks, whereas a full column, together with W , has $k + 2$; the latter is the full star with common $(k - 1)$ -set $T = W \setminus \{p\}$. Thus (4) is the only equality type. Its two $\equiv$ -components have the distinct sizes $k + 2$ and $k + 1$, so the common set T and the remaining point p are uniquely recoverable from the family. This gives $(k + 2)\binom{2k+1}{k-1}$ families. $\square$

3 From an endomorphism to equal fibres

Put

$$N = |\Omega| = \binom{2k+1}{k}.$$

The following lemmas reduce a noninjective endomorphism to two divisibility conditions and a transversal identity.

Lemma 3.1 (Minimum-rank retraction). *If G_k has a non-automorphic endomorphism, then it has a retraction $r : G_k \rightarrow H$ onto a proper core H . All fibres of r have a common size $s > 1$. If $q = |H|$, then*

$$sq = N, \quad s \mid N. \quad (5)$$

Moreover, for every fibre $F_y = r^{-1}(y)$ and every $\gamma \in S_{2k+1}$,

$$|F_y \cap \gamma H| = 1. \quad (6)$$

Proof. Choose an endomorphism ϕ of minimum rank and put $H = G_k[\phi(\Omega)]$, the subgraph induced by its vertex image. The restriction $\phi|_H$ is injective, for otherwise ϕ^2 has smaller rank. Since H is finite, this restriction is an automorphism. Composing ϕ with its inverse on H gives a retraction

$$r = (\phi|_H)^{-1} \circ \phi : G_k \rightarrow H.$$

If H had a noninjective endomorphism, composing it with r would produce an endomorphism of G_k with smaller rank. Thus H is a core.

For $y \in H$ and $\gamma \in S_{2k+1}$, the map $r \circ \gamma|_H : H \rightarrow H$ is an endomorphism of the core H , hence an automorphism. Exactly one point of γH therefore lies in the fibre F_y , proving (6).

Average (6) over γ . Since the point-permutation group is transitive on Ω , each vertex belongs to exactly $|S_{2k+1}|q/N$ translates γH . Hence

$$|S_{2k+1}| = \sum_{\gamma} |F_y \cap \gamma H| = |F_y| \frac{|S_{2k+1}|q}{N}.$$

Thus $|F_y| = N/q$, independently of y , and (5) follows. This also gives a direct proof, in the present setting, of the equal-fibre phenomenon for vertex-transitive graphs [7, 10]. $\square$

Lemma 3.2 (Divisibility by $2k + 3$). *Under the hypotheses of Lemma 3.1,*

$$s\alpha(H) = 2k + 3. \quad (7)$$

In particular,

$$s \mid 2k + 3. \quad (8)$$

Proof. If I is independent in H , then $r^{-1}(I)$ is independent in G_k ; equal fibres and Proposition 2.1 give

$$\alpha(H) \leq \frac{2k+3}{s}. \quad (9)$$

The inclusion $H \hookrightarrow G_k$ and the retraction give homomorphisms in both directions, so $\chi_f(H) = \chi_f(G_k)$. A vertex-transitive graph G satisfies $\chi_f(G) = |V(G)|/\alpha(G)$, while every graph Y satisfies $\chi_f(Y) \geq |V(Y)|/\alpha(Y)$ [11, Chapters 1–3]. Consequently

$$\frac{N}{2k+3} = \chi_f(G_k) = \chi_f(H) \geq \frac{q}{\alpha(H)} = \frac{N/s}{\alpha(H)}.$$

This reverses (9), proving (7) and hence (8). $\square$

4 Spectral transversals in the Johnson scheme

Let A_i be the matrix of Johnson distance i , so $A_i(A, B) = 1$ precisely when $d(A, B) = i$. Let ρ denote the permutation representation of S_{2k+1} on $\mathbb{R}^\Omega$. Its multiplicity-free decomposition is

$$\mathbb{R}^\Omega = V_0 \oplus V_1 \oplus \cdots \oplus V_k,$$

where $V_j \cong S^{(2k+1-j, j)}$. Write $\rho_j = \rho|_{V_j}$ and let E_j be orthogonal projection onto V_j . This is the standard Johnson association scheme; see, for example, [2, 6].

Lemma 4.1 (Spectral-transversal lemma). *Let F be a fibre and let H be the core image in Lemma 3.1. For every $j \geq 1$,*

$$\|E_j \mathbf{1}_F\|^2 \|E_j \mathbf{1}_H\|^2 = 0. \quad (10)$$

Proof. By (6), for all $\gamma \in S_{2k+1}$,

$$\langle \mathbf{1}_F, \rho(\gamma) \mathbf{1}_H \rangle = 1.$$

As $|F||H| = sq = N$, removing the constant components gives

$$\left\langle \mathbf{1}_F - \frac{s}{N} \mathbf{1}, \rho(\gamma) \left(\mathbf{1}_H - \frac{q}{N} \mathbf{1} \right) \right\rangle = 0.$$

After decomposing into $V_1, \dots, V_k$, we may complexify; the real Specht modules here are absolutely irreducible. Schur orthogonality then says that the matrix-coefficient spaces belonging to distinct irreducibles are orthogonal. Since their sum is the zero function, every individual function

$$\gamma \mapsto \langle E_j \mathbf{1}_F, \rho_j(\gamma) E_j \mathbf{1}_H \rangle$$

vanishes identically. If $E_j \mathbf{1}_H \neq 0$, its orbit spans the irreducible module V_j , forcing $E_j \mathbf{1}_F = 0$. This is exactly (10). $\square$

4.1 Endpoint eigenvalues and fibre moments

Use the first-eigenmatrix convention

$$A_i E_j = P_j(i) E_j.$$

The source [1, Section 5.1 and Theorem 5.4] uses the opposite order for the two indices. With our convention, the Eberlein polynomials are

$$P_j(i) = \sum_{r=0}^i (-1)^{i-r} \binom{k-r}{i-r} \binom{k-j}{r} \binom{k+1+r-j}{r}. \quad (11)$$

Put

$$K = k + 1, \quad t = K - j, \quad \epsilon = (-1)^j. \quad (12)$$

Specializing (11), and using $A_1 A_k = 2A_{k-1} + (K-1)A_k$ for the middle endpoint, gives

$$\boxed{P_j(1) = t^2 - K, \quad P_j(k-1) = \epsilon \frac{t(t^2 - 2K + 1)}{2}, \quad P_j(k) = \epsilon t.} \quad (13)$$

Let v_i denote the valency of relation i . At the three relevant distances,

$$v_1 = K(K-1), \quad v_{k-1} = \frac{K(K-1)^2}{2}, \quad v_k = K. \quad (14)$$

Each fibre is independent, so its distinct pairs have only distances $1, k-1, k$. Let x, y, z be the numbers of unordered pairs at these three distances and normalize

$$p_1 = \frac{2x}{s(s-1)}, \quad p_{k-1} = \frac{2y}{s(s-1)}, \quad p_k = \frac{2z}{s(s-1)}; \quad p_1 + p_{k-1} + p_k = 1. \quad (15)$$

Here $\mathbf{1}_F^\top A_1 \mathbf{1}_F = 2x$, and similarly at distances $k-1$ and k ; this accounts for the factor 2. With $\mu_j = \dim V_j$, the primitive-idempotent expansion

$$E_j = \frac{\mu_j}{N} \sum_{i=0}^k \frac{P_j(i)}{v_i} A_i$$

shows that

$$\|E_j \mathbf{1}_F\|^2 = \frac{\mu_j}{N} \eta_j, \quad (16)$$

where, writing $e_j = \eta_j/s$,

$$e_j = 1 + (s-1)(a_j p_1 + b_j p_{k-1} + c_j p_k) \quad (17)$$

with

$$a_j = \frac{t^2 - K}{K(K-1)}, \quad b_j = \frac{\epsilon t(t^2 - 2K + 1)}{K(K-1)^2}, \quad c_j = \frac{\epsilon t}{K}. \quad (18)$$

In particular, every η_j and e_j is nonnegative.

For the core image define

$$h_j = \|E_j \mathbf{1}_H\|^2, \quad R_i = \mathbf{1}_H^\top A_i \mathbf{1}_H = \sum_{j=0}^k P_j(i) h_j \geq 0. \quad (19)$$

Since $sq = N$,

$$h_0 = \frac{q^2}{N} = \frac{q}{s}, \quad \sum_{j \geq 1} h_j = \frac{q(s-1)}{s}. \quad (20)$$

Lemma 4.1 and (16) give the complementarity condition

$$\eta_j h_j = 0 \quad (1 \leq j \leq k). \quad (21)$$

Lemma 4.2 (The first module). *Every fibre satisfies $\eta_1 > 0$. Consequently*

$$h_1 = 0. \quad (22)$$

Proof. If $E_1 \mathbf{1}_F = 0$, then all point degrees in F are equal, with common value $sk/(2k+1)$. Since $\gcd(k, 2k+1) = 1$, integrality gives $2k+1 \mid s$. But $s \mid 2k+3$ by Lemma 3.2, and $\gcd(2k+1, 2k+3) = 1$, a contradiction to $s > 1$. Thus $\eta_1 > 0$, and (22) follows from (21). $\square$

Remark 4.3 (A design consequence). For $j = 2$, the three normalized endpoint eigenvalues are

$$\frac{k-3}{k+1}, \quad \frac{(k-1)(k-4)}{k(k+1)}, \quad \frac{k-1}{k+1}.$$

They are nonnegative for $k \geq 4$, so the defining moment formula gives $\eta_2 \geq s$ and hence $h_2 = 0$. The point-block and pair-block incidence matrices have column spaces $V_0 \oplus V_1$ and $V_0 \oplus V_1 \oplus V_2$, respectively. Hence vanishing of the first two nonconstant Johnson components is equivalent to constant point and pair degrees, that is, to the 2-design conditions. Double counting incidences and point-pairs gives

$$r_H = \frac{kq}{2k+1}, \quad \lambda_H = \frac{(k-1)q}{2(2k+1)}.$$

This observation is not needed in the proof of Theorem 1.1.

5 Proper divisors of $2k+3$

Set

$$m = 2k+3 = 2K+1. \quad (23)$$

Suppose $1 < s < m$. Because $s \mid m$ and m is odd,

$$3 \leq s \leq \frac{2K+1}{3} \leq K. \quad (24)$$

We now exclude these values of s .

Lemma 5.1 (Positivity for even modules). *If $1 \leq j \leq k$ is even and $3 \leq s \leq K$, then*

$$e_j \geq \frac{K-s+1}{K} > 0. \quad (25)$$

Proof. Here $\epsilon = 1$. The three coefficient shifts are

$$a_j + \frac{1}{K} = \frac{t^2 - 1}{K(K-1)} \geq 0, \quad (26)$$

$$b_j + \frac{1}{K} = \frac{t^3 - (2K-1)t + (K-1)^2}{K(K-1)^2} > 0, \quad (27)$$

$$c_j + \frac{1}{K} = \frac{t+1}{K} > 0. \quad (28)$$

For the strict sign in (27), write the numerator as $\Psi_K(t)$. If $t = 1$, then $\Psi_K(1) = (K-1)(K-3) > 0$. If $t \geq 2$, then $K \geq t+1$, $\Psi_K(t)$ is increasing as a function of integer $K \geq t+1$, and $\Psi_{t+1}(t) = t(t^2 - t - 1) > 0$. Using $p_1 + p_{k-1} + p_k = 1$ in (17), we obtain the affine identity

$$e_j - \frac{K-s+1}{K} = (s-1) \left[\left(a_j + \frac{1}{K} \right) p_1 + \left(b_j + \frac{1}{K} \right) p_{k-1} + \left(c_j + \frac{1}{K} \right) p_k \right] \geq 0. \quad (29)$$

$\square$

Lemma 5.2 (Zero criterion for odd modules). *Suppose $3 \leq s \leq K$ and $1 \leq j \leq k$ is odd. Put $t = K - j$ and define*

$$D_j = K - (s - 1)t.$$

If $D_j \geq 0$, then $e_j > 0$. Consequently

$$e_j = 0 \implies j \text{ is odd and } D_j \leq -1. \quad (30)$$

Proof. Since $\epsilon = -1$,

$$A_t := a_j - c_j = \frac{(t-1)(t+K)}{K(K-1)} \geq 0, \quad C_t := b_j - c_j = \frac{t(K^2 - t^2)}{K(K-1)^2} > 0. \quad (31)$$

Eliminating p_k gives

$$e_j - \frac{D_j}{K} = (s-1)(A_t p_1 + C_t p_{k-1}). \quad (32)$$

If $D_j > 0$, then (32) gives $e_j \geq D_j/K > 0$. Suppose $D_j = 0$. Then $(s-1)t = K$ and $3 \leq s \leq K$, so

$$2 \leq t \leq K/2, \quad j-1 = K-t-1 > 0.$$

With

$$\lambda_t = \frac{(t-1)(t+K)}{(K-2)(2K-1)},$$

simplification gives

$$e_j - \lambda_t e_1 - \lambda_t \frac{j-1}{t} = \frac{(K+t)(K-t-1)(t(K-2)+1)}{t(K-2)(K-1)^2} p_{k-1} \geq 0. \quad (33)$$

The coefficient λ_t is chosen to eliminate the p_1 term. Moreover, $\lambda_t(j-1)/t$ is positive. Since $e_1 \geq 0$, this proves $e_j > 0$. Together with Lemma 5.1, this yields (30); D_j is integral, so $D_j < 0$ means $D_j \leq -1$. $\square$

Proposition 5.3. *No proper divisor s of m can be the common fibre size of a proper core retraction of G_k .*

Proof. By (21), the nonconstant support of (h_j) is contained in the zero set of (e_j) . Lemmas 5.1 and 5.2 show that every supported index is odd and has $D_j \leq -1$. Let

$$W = \sum_{j \geq 1} h_j = \frac{q(s-1)}{s}, \quad w_j = \frac{h_j}{W}.$$

For supported odd j , (13) gives $P_j(k) = -t$. Including the constant contribution $P_0(k)h_0 = Kq/s$, we obtain

$$\frac{s}{q} R_k = K - (s-1) \sum_j w_j t = \sum_j w_j (K - (s-1)t) = \sum_j w_j D_j \leq -1. \quad (34)$$

Thus $R_k \leq -q/s < 0$, contradicting the nonnegativity in (19). $\square$

6 The case $s = 2k + 3$

Proposition 6.1. *The value $s = 2k + 3$ cannot be the common fibre size of a proper core retraction of G_k .*

Assume throughout this section that

$$s = m = 2K + 1. \quad (35)$$

For $t = K - j$ and $\epsilon = (-1)^j$, write $e_{\epsilon,t} := e_j$. Equation (17) becomes the affine function

$$e_{\epsilon,t} = 1 + \frac{2(t^2 - K)}{K - 1}p_1 + \frac{2\epsilon t(t^2 - 2K + 1)}{(K - 1)^2}p_{k-1} + 2\epsilon tp_k. \quad (36)$$

The required positivity follows from interpolation among three endpoint functions.

Lemma 6.2 (Endpoint interpolation). *Under the assumption $s = 2K + 1$, suppose $K \geq 5$.*

1. *If K is odd, set*

$$B_1 = e_{-,K-1} = e_1, \quad B_{k-1} = e_{-,2} = e_{k-1}, \quad B_k = e_{+,1} = e_k.$$

On the parity grids

$$(-, t), \quad t = 2, 4, \dots, K - 1, \quad (+, t), \quad t = 1, 3, \dots, K - 2,$$

every $e_{\epsilon,t}$ is a nonnegative linear combination of B_1, B_{k-1}, B_k . Away from the three displayed endpoints, the coefficient of B_1 is positive.

2. *If K is even, the same conclusion holds with*

$$B_1 = e_{-,K-1} = e_1, \quad B_{k-1} = e_{+,2} = e_{k-1}, \quad B_k = e_{-,1} = e_k$$

on the grids

$$(-, t), \quad t = 1, 3, \dots, K - 1, \quad (+, t), \quad t = 2, 4, \dots, K - 2.$$

Proof. Solving the affine 3×3 systems gives the factored coefficients in Appendix A. Their denominators are positive for $K \geq 5$, and the factors in their numerators give the stated signs on the two parity grids. $\square$

Proof of Proposition 6.1. First suppose that K is odd. Since $B_1 = e_1 > 0$ by Lemma 4.2, Lemma 6.2 shows that every non-endpoint e_j is positive. Complementarity (21) and $h_1 = 0$ therefore confine the nonconstant core support to $j = k - 1, k$.

At these endpoints,

	$P_j(1)$	$P_j(k)$
$j = k - 1$	$4 - K$	-2
$j = k$	$1 - K$	1

Using $h_{k-1} + h_k = q(s - 1)/s$, $s - 1 = 2K$, and the constant contribution $P_0(1) + P_0(k) = K^2$, we obtain

$$R_1 + R_k = \frac{q}{s}K(4 - K) < 0. \quad (37)$$

This contradicts $R_1, R_k \geq 0$ for every odd $K \geq 5$.

Now suppose that K is even. The same argument again leaves only $j = k - 1, k$. Here

$$P_{k-1}(1) = 4 - K, \quad P_k(1) + 3 = 4 - K.$$

Together with $P_0(1) = K(K - 1)$ and $s - 1 = 2K$, this gives

$$R_1 + 3h_k = \frac{q}{s}K(7 - K). \quad (38)$$

For even $K \geq 8$ the right side is negative and the left side is nonnegative. The only smaller even value is $K = 6$ ($k = 5$), but then $m = 13$ does not divide $N = \binom{11}{5} = 462$, so this fibre size is not a candidate. $\square$

7 Proof of the main theorem and consequences

Proof of Theorem 1.1. Suppose that G_k has a non-automorphic endomorphism. By Lemma 3.1, a minimum-rank choice gives a proper core retraction with common fibre size $s > 1$. Lemmas 3.1 and 3.2 give

$$s \mid \binom{2k+1}{k}, \quad s \mid 2k+3.$$

If $s < 2k + 3$, Proposition 5.3 gives a contradiction. If $s = 2k + 3$, Proposition 6.1 gives a contradiction. Hence every endomorphism is injective. On a finite graph it is therefore bijective; a bijective self-homomorphism permutes the finite edge set injectively and hence is an automorphism. Finally (2) identifies the group as S_{2k+1} . $\square$

Corollary 7.1. *For every odd integer $n \geq 9$, the almost-uniform qualitative independence hypergraph $3\text{-AUQI}(n, 2)$ is a core.*

Proof. Write $n = 2k + 1$, so $k \geq 4$. Theorem 1.1 says that the merged Johnson graph G_k is a core. The transfer result of Thomas and Akhtar [12, Lemma 7 and Proposition 7] then gives the stated hypergraph conclusion. $\square$

A Coefficients for endpoint interpolation

This appendix records the coefficients used in Lemma 6.2. They are obtained by substituting $p_k = 1 - p_1 - p_{k-1}$ into (36) and solving in the affine basis $1, p_1, p_{k-1}$; direct substitution verifies the identities below.

Odd K

With the endpoint functions from part (1) of Lemma 6.2, the negative-parity functions satisfy

$$e_{-,t} = \alpha_O^- B_1 + \beta_O^- B_{k-1} + \gamma_O^- B_k, \quad (39)$$

where

$$\begin{aligned} \alpha_O^- &= \frac{(t-2)(t+1)[K(K-t-1)+4t-3]}{K(K-3)(4K-7)}, \\ \beta_O^- &= \frac{(t+1)(K-t-1)[t(2K-5)+3]}{3(K-3)(4K-7)}, \\ \gamma_O^- &= \frac{(t-2)(K-t-1)[4t(K-1)+3]}{3K(4K-7)}. \end{aligned}$$

The positive-parity functions satisfy

$$e_{+,t} = \alpha_O^+ B_1 + \beta_O^+ B_{k-1} + \gamma_O^+ B_k, \quad (40)$$

where

$$\begin{aligned} \alpha_O^+ &= \frac{(t-1)(t+2)[K(K-1) + t(K-4) - 3]}{K(K-3)(4K-7)}, \\ \beta_O^+ &= \frac{(t-1)(K+t-1)[t(2K-5) - 3]}{3(K-3)(4K-7)}, \\ \gamma_O^+ &= \frac{(t+2)(K+t-1)[4t(K-1) - 3]}{3K(4K-7)}. \end{aligned}$$

For odd $K \geq 5$, the denominators are positive. On the grids in Lemma 6.2, every numerator is nonnegative, and $\alpha_O^\pm$ is positive away from the three endpoints.

Even K

With the endpoint functions from part (2) of the lemma, the negative-parity functions satisfy

$$e_{-,t} = \alpha_E^- B_1 + \beta_E^- B_{k-1} + \gamma_E^- B_k, \quad (41)$$

where

$$\begin{aligned} \alpha_E^- &= \frac{(t-1)(t+2)[K(K-t+1) + 2t-1]}{(K-2)(K+1)(4K-3)}, \\ \beta_E^- &= \frac{(t-1)(K-t-1)[t(2K-1) + 1]}{3(K+1)(4K-3)}, \\ \gamma_E^- &= \frac{(t+2)(K-t-1)[4t(K-2) + 5]}{3(K-2)(4K-3)}. \end{aligned}$$

The positive-parity functions satisfy

$$e_{+,t} = \alpha_E^+ B_1 + \beta_E^+ B_{k-1} + \gamma_E^+ B_k, \quad (42)$$

where

$$\begin{aligned} \alpha_E^+ &= \frac{(t-2)(t+1)[K(K+t+1) - 2t-1]}{(K-2)(K+1)(4K-3)}, \\ \beta_E^+ &= \frac{(t+1)(K+t-1)[t(2K-1) - 1]}{3(K+1)(4K-3)}, \\ \gamma_E^+ &= \frac{(t-2)(K+t-1)[4t(K-2) - 5]}{3(K-2)(4K-3)}. \end{aligned}$$

For even $K \geq 6$, the same sign conclusions hold on the corresponding parity grids.

Code availability

The supplement contains two exact-arithmetic Python scripts. They check selected identities symbolically, reconstruct the interpolation coefficients, and run the finite-range tests listed in the supplementary README; they do not mechanize the combinatorial or representation-theoretic arguments. The scripts were tested with Python 3.9 and SymPy 1.14 and can be run with `make check`.

Declaration on AI-assisted tools

The author used OpenAI language models for conjecture exploration, symbolic calculation, proof review, and prose editing, and reviewed all resulting material. The manuscript is licensed under CC BY 4.0; the accompanying Python scripts are licensed under the MIT License.

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